

GUI-Based Platform for Transmission Line Evaluation and Power System Insights

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ABSTRACT The evaluation of transmission line parameters plays a vital role in selecting the appropriate conductor, calculating losses and efficiency, and ensures smooth power delivery. However, existing tools and methods for evaluating transmission line parameters often lack accuracy, They offer limited customizability, and are too complex for manual computation. This paper presents a Graphical User Interface (GUI)-based platform developed in MATLAB for the evaluation and visualization of transmission lines and power system analysis. The proposed tool collects user-provided data on the dimensions and structure of the transmission line and generates real-time computation and visualization of various transmission line characteristics. By offering an interactive and simulation-based environment, the tool enhances the user's understanding of transmission line behavior and supports more effective design decisions.

INDEX TERMS Transmission Line Parameters, MATLAB GUI, Power System Analysis, Pi-Model, Efficiency Calculation, Loss Estimation, Ampacity, Severity Index

I. INTRODUCTION

In this fast-paced world, the generation of power is increasing, and this sets the need for power systems everywhere. Power systems rely on accurate transmission and distribution networks, so their efficiency plays a huge role. The planning and calculation of constructing such transmission lines will reduce losses and ensure stability. Conventional analytical methods require long, complex manual calculations. To resolve this issue, a good software tool to simulate such real-time structures and contractions is needed. Power System Evaluation frameworks aim to bring all these under one environment. With the rapid integration of renewable sources and distributed generation, transmission lines now operate under diverse dynamic conditions, demanding more adaptive analytical tools [6]. MATLAB and Simulink environments have become powerful educational and research platforms for simulating transient and steady-state behaviors [7]. Existing research, such as Koç and Aydoğmus [1] demonstrated that MATLAB-based GUIs can

effectively simplify power system education and fault simulation analysis.

A. MOTIVATION

The efficiency of the transmission lines depends on their parameters. Resistance and inductance influence power loss and voltage regulation. Capacitance and conductance affect charging currents and leakage. The calculation of R, L, and C values is one of the first steps in designing a transmission line. Parameter estimation of transmission lines has been a long-standing focus area, as highlighted by Shaikh et al. [2], who proposed an optimized approach for calculating R, L, and C values. Different line lengths (short, medium, long) require specific models (Nominal-T, Nominal- π , or long line equations) because the importance of line parameters like resistance, inductance, and especially capacitance changes with distance. Even small deviations in calculated parameters can significantly affect system performance [3]. A Modern analysis tool that can calculate these parameters to assess voltage regulation, efficiency, and losses boosts the system

performance. Additionally, ampacity assessment supports the safe operation of conductors under varying load and temperature conditions, a feature rarely embedded in earlier GUI tools [10].

Traditional tools or MATLAB scripts require manual input and coding skills. Although several simulation tools exist, many lack user interactivity or require advanced coding knowledge. This limits the accessibility for students and early-stage researchers [8]. Earlier computational tools, such as the calculator developed by Bahadoorsingh et al. [4], focused on automating parameter evaluation but lacked graphical visualization. A Graphical User Interface (GUI) provides a simple, interactive, and customizable approach. Users can visualize power flow, losses, and ampacity instantly. The concept of GUI-based analysis in power systems dates back to early tools such as that by Yu et al. [5], validating the enduring relevance of graphical engineering platforms. GUI-based environments help users interpret electrical quantities such as voltage profiles and efficiency through direct visualization, improving conceptual understanding [9]. This tool benefits students, researchers, and engineers by combining simulation, computation, and visualization in one application.

Thus, this paper provides a GUI-Based MATLAB software tool to get the transmission line parameters and visualizations of various aspects of parameters and losses by getting the input data from the user.

B. IDEA FORMULATION

The plan is to implement a GUI for the computation of transmission line parameters, efficiency, regulation, analyse the efficiency before and after the application of different kinds of losses which are due to various other factors, compute ampacity score and severity index which would be very useful while designing the circuit for the transmission line.

Given below table explains the key aspects which were carried out and executed during the course of this project.

TABLE I
KEY ASPECTS OR ELEMENTS OF THIS PROJECT

Section/Element	Description
Transmission Line Parameter Evaluation	This part of GUI takes in inputs like conductor specifications, operating temperature, tower configuration, etc and evaluates the resistance, inductance and capacitance under the given conditions
Transmission Line Efficiency and Regulation	This section will take the necessary inputs and calculate the efficiency and regulation based on the short, medium or long transmission line model chosen.
Computation of different losses and their contribution to the overall efficiency	User will be allowed to choose from various losses provided, and based on the losses chosen, the overall power will be computed and thus efficiency value is also affected from that of the normal efficiency which was calculated in the previous section.
Ampacity Score Calculation	Ampacity (current capacity) is a very important factor while sizing the wire. A weighted score will be calculated telling how safe the provided wire sizing inputs are, and comments will be given.
Fault Efficiency, Severity Index	Efficiency at the time of fault is noted, and severity index is calculated by the deviation of efficiency from normal working conditions.

II. WORKFLOW AND SIGNIFICANCE

Each transmission line is unique by its parameters, R, L and C. These parametric values change and behave differently based on the physical and environmental conditions provided. Power flow, efficiency, and regulation all the other subsequent sections are highly dependent on this and thus forming the base of this project. Efficiency and Regulation are the most important factors, and decides how good a line is suited for the application demanded. From a design perspective these are very crucial in validating the performance of the overall transmission line.

The user is given the freedom to choose whatever losses that need to be considered for his situation or design. This section helps to find out which loss is dominating and whether any further improvements can be made. Based on the entered environmental and physical conditions, the capacity of current is computed for that particular wire geometry. This score helps in telling whether the chosen conductor size is good enough for the working conditions of the transmission line. This helps in wire selection and thermal safety.

During faults, the system behaves differently. Higher the calculated severity index, means the system is adversely affected by the fault introduced. This helps out the engineers by giving clear details on the worst-case efficiency which is very useful for them to plan the fanout/load operations accordingly.

The [13] discusses a few fundamental formulas which are essential for the calculation of ampacity current.

$$h = 3 + 5.7v \quad (i)$$

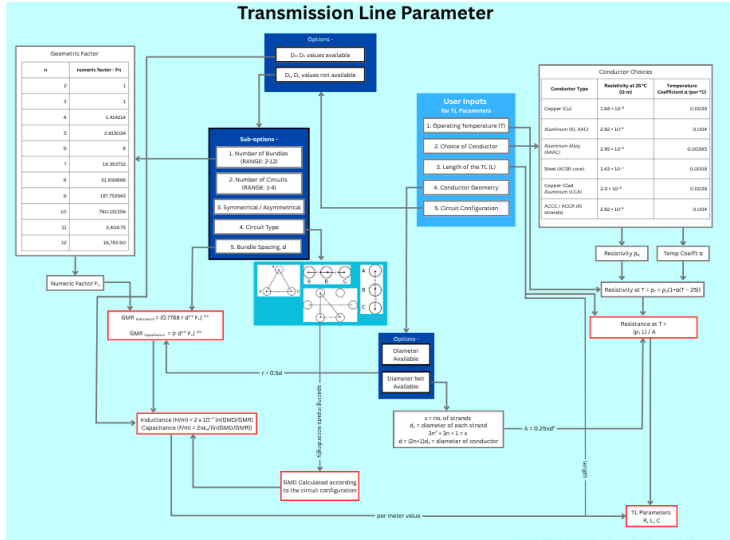
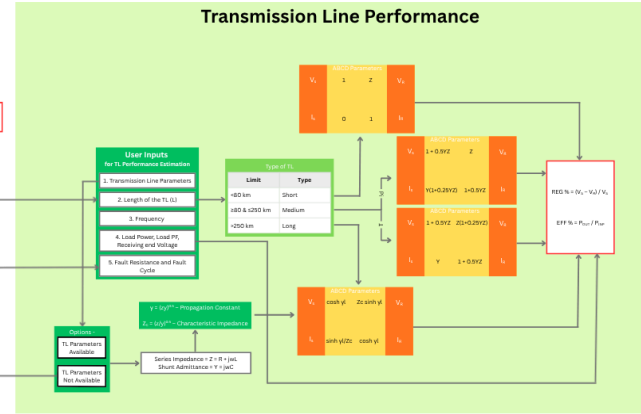


FIGURE 1: Workflow of the Proposed GUI



III. THEORETICAL ASPECTS OF THE WORK

This section discusses the basic principles and ideologies involved while implementing and executing this work.

A. LITERATURE REVIEW

The conductor heats up when current flows through it. This heat balance is disturbed due to weather factors such as wind, ambient temperature, and solar irradiance. Static ratings use worst-case assumptions. Dynamic weather can offer extra capacity compared to conservative static ratings [11]. Other concerning environmental factors include icing and ice coating. Ice coating, generally, is known to add extra weight and thermal insulation. The ice prediction work shows icing is complex and needs multi-sensor or a time-series treatment [12]. These papers lead to a conclusion that for ampacity calculation, the environmental parameters need to be accounted for and be given more weightage.

(i) is the convective heat transfer coefficient, where v is the wind speed (in m/s). Also the total thermal

dissipation per meter is given by,

$$Q_{total} = Q_{conv} + Q_{rad} \quad (ii)$$

Where $Q_{conv} = h \cdot p(T_{allowable} - T_{ambient})$ and,

$$Q_{rad} = \varepsilon \cdot \sigma \cdot p(T_{allowable}^4 - T_{ambient}^4)$$

From (i), (ii) the $I_{ampacity}$ can be calculated [13]. Ampacity score calculation is discussed in the next section.

R,L,C calculation of a transmission line for a given set of conductor attributes is done by MathWorks line parameter calculator [14]. It computes the exact R,L,C matrices, supports bundled conductors, ground wires, specifying exact conductor positions, sag and multiple

conductor types [14]. Important gaps that need to be addressed include, ampacity solver according [13] standards. Also weather inputs, corona loss can be added. Ideally a full operational electrical tool should connect the electrical parameters calculated to thermal and environmental models.

Faults are of many types, which are classified on the basis of the terminal between which they occur. Those include SLG, LL, LLG, LLL, LLLG. SLG is the most common but also less severe, while the case of LLL, LLLG is vice-versa [15]. A similar pattern can also be noticed in the simulink results obtained in the later section.

Corona loss is generally ignored in the power flow and efficiency calculations due to its negligible values (much lower compared to the other losses). But it is shown that corona loss of 500 kV and above line is very sensitive and accounts for a high proportion of the total loss, thus making it mandatory to include it in power flow and efficiency calculations for more reliability [16]. This loss can be computed by two different formulas namely, Peek (iii) and Peterson (iv) formulas [16]. Corona power loss is given by,

$$P_c = \frac{244}{\delta} (f + 25) (E_n - E_0)^2 \frac{\sqrt{GMR}}{\sqrt{GMD}} 10^{-5} \quad (\text{KW/Km/Ph}) \quad (\text{iii})$$

Where, δ is the air density factor; E_0 is the disruptive critical voltage per phase; E_n – r.m.s phase voltage in kV

$$P_c = 2.1 f W \frac{E_n^2}{(\log_{10} \frac{GMD}{GMR})^2} \times 10^{-5} \quad (\text{iv})$$

W is a factor called corona loss function which varies with the ratio E_n/E_0 . E_0 is calculated by using the below expression.

$$E_0 = G_0 m_0 r \delta^{\frac{2}{3}} \ln \frac{GMD}{GMR} \text{ V/ph} \quad (\text{v})$$

Where, G_0 is the maximum value of disruptive critical voltage gradient in V/m; m_0 = irregularity factor

Though Peek's formula is not as complicated or complex as Peterson's formula, it is standard, simple

and is a good baseline. Thus this project implements the Peek's formula at present to compute the corona loss.

B. DERIVED FORMULAS

There were several cases at which direct formulations were not available and only a basic idea or concept about that particular case was available. Such formulas are covered in this section.

The basic GMR, GMD formulas are discussed in brief by the workflow diagram in the previous page. But for a tower configuration like delta (triangular) when the number of circuits = 2, it can be written as follows (see figure 2)

For symmetric circuit spacing ($x=y=z$),

$$\text{height, } h = \frac{\sqrt{3}x}{2} \quad (\because \sin\theta = \frac{h}{x}, \theta = 60^\circ)$$

$$D_{AB} = \sqrt[4]{x^2 \cdot \sqrt{h^2 + (w-u)^2} \cdot \sqrt{x^2 + w^2 + xw}}$$

$$D_{BC} = \sqrt[4]{y^2 \cdot u(u+2y)}$$

$$D_{CA} = \sqrt[4]{z^2 \cdot \sqrt{h^2 + (w-u)^2} \cdot \sqrt{z^2 + w^2 - zw}}$$

For asymmetric circuit spacing ($x \neq y \neq z$)

Using heron's formula, $s = \frac{x+y+z}{2}$

$$A = \sqrt{s \cdot (s-x) \cdot (s-y) \cdot (s-z)}$$

$$\text{height, } h = \frac{2xA}{y}$$

$$D_{AB} = \sqrt[4]{x^2 \cdot \sqrt{h^2 + (w-u)^2} \cdot \sqrt{x^2 + w^2 + 2xw \cos(\sin^{-1} \frac{h}{x})}}$$

$$D_{BC} = \sqrt[4]{y^2 \cdot u(u+2y)}$$

$$D_{AB} = \sqrt[4]{z^2 \cdot \sqrt{h^2 + (w-u)^2} \cdot \sqrt{z^2 + w^2 - 2zw \cos(\sin^{-1} \frac{h}{x})}}$$

$$GMD = \sqrt[3]{D_{AB} D_{BC} D_{CA}}$$

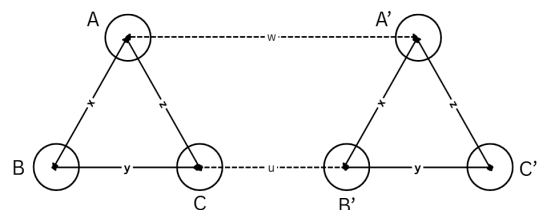


FIGURE 2: Showing double circuit delta conductor spacing

For ampacity weighted-score calculation the below algorithm is used.

The worst-case current, I_{MAX} is,

$$I_{max} = \max(|I_{send}|, |I_{receive}|)$$

$$\text{Utilization factor, } U = \frac{I_{max}}{I_{ampacity}}$$

Base score function is given by the below piece-wise function, S_{base}

$$\begin{aligned} S_{base} &= 100, U \leq 0.7 \\ &= 100 - \frac{U-0.7}{0.3} \cdot 40, 0.7 < U \leq 1.0 \\ &= 60 - \frac{U-1}{0.2} \cdot 40, 1 < U \leq 1.2 \\ &= \max(0, 20 - \frac{U-1.2}{1} \cdot 20), U > 1.2 \end{aligned}$$

$$I_{imbalance} = \frac{|I_{send} - I_{receive}|}{I_{max}}$$

Score after imbalance correction,

$$S_{final} = S_{base} \cdot (1 - \min(0.15, 0.15 \cdot I_{imbalance}))$$

Here the ampacity score, higher the safety

The severity index for a particular fault is given by,

$$\text{Severity Index, } SI = 1 - \frac{\eta_{fault}}{\eta_{normal}}$$

Higher the severity, more severe the fault

IV. RESULTS AND DISCUSSIONS

A. RESISTANCE, INDUCTANCE & CAPACITANCE

Table 2: Showing inductance value (taken from GUI) for a conductor of diameter = 0.024m for various tower configurations

Tower config	No of circuits	L (mH/Km)		
		n=2	n=3	n=4
Delta	1 (4m)	0.9866	0.8680	0.7541
	2 (4m,6m)	0.4770	0.4177	0.3607

Horizontal	1 (4m)	1.0903	0.9718	0.8578
	2 (4m,6m)	0.5555	0.4962	0.4392
Vertical	1 (4m)	1.0328	0.9142	0.8003
	2 (4m,6m)	0.7396	0.6211	0.5071
Hexagonal	2 (sym)	0.9886	0.8701	0.7561
	2 (asym)	0.4389	0.3796	0.3226

Table 3: Showing capacitance value (taken from GUI) for a conductor of diameter = 0.024m for various tower configurations

Tower config	No of circuits	C (nF/Km)		
		n=2	n=3	n=4
Delta	1 (4m)	11.571	13.069	15.004
	2 (4m,6m)	10.9670	12.303	14.003
Horizontal	1 (4m)	10.444	11.650	13.162
	2 (4m,6m)	9.553	10.552	11.778
Vertical	1 (4m)	11.040	12.396	14.124
	2 (4m,6m)	10.370	11.557	13.044
Hexagonal	2 (sym)	10.094	11.216	12.611
	2 (asym)	11.882	13.467	15.530

The calculated inductance falls in the range of 0.8-1.1 mH/km while the capacitance falls in the range of 10-14nF/km. These values are nominal and valid enough for mid-sized overhead cables.

Table 4: Showing resistance value (taken from GUI) for a conductor of diameter 25mm, 35mm for Cu, Al conductors (L=100Km)

Conductor	Cond dia (mm)	R (Ω)		
		T ₁ = 5°C	T ₂ = 25°C	T ₃ = 70°C
Cu	25	3.22854	3.50396	4.1263

	35	1.64722	1.78773	2.1089
Al	25	4.9774	5.3985	6.346
	35	2.5395	2.7544	3.2377

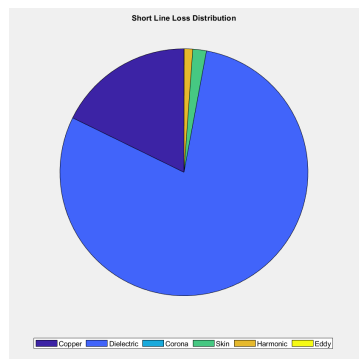
From the above table, R values are also reasonable for a 100 Km transmission line. It can be noted that from 5°C - 25°C, R increases around 8.5%. Similarly from 25°C - 70°C, R increases around 18%. Overall increase in R is around 26%. Aluminum has 1.54 times the resistance offered by that of Copper, which is also validated as the resistivity of Aluminum is greater than Copper. These results also prove that GUI is working correctly.

B. LOSS DISTRIBUTION

Table 5: Input values used in GUI to plot loss distribution chart and compute efficiency (normal) and efficiency (after loss) for short, medium and long transmission line models

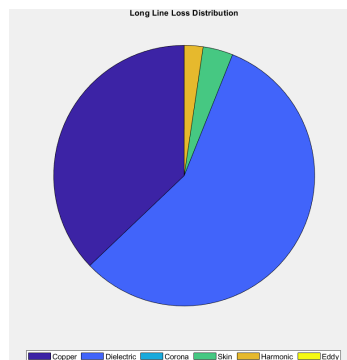
Parameters	Values
R	0.12 ohm/km
L	1.1 mH/km
C	1 nF/km
THD	5%
$\tan \delta$	0.0156
m_0	0.92
P	780 torr
T_c	45°C

Short Transmission line model (50 km)



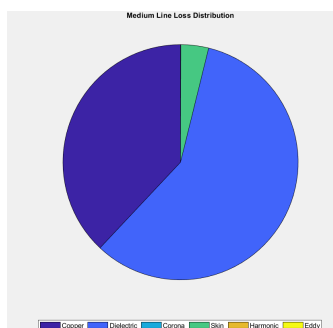
$\eta_{\text{no-loss}}$	99.61
$\eta_{\text{after-loss}}$	97.19

Long Transmission line model (300 km)



$\eta_{\text{no-loss}}$	97.73
$\eta_{\text{after-loss}}$	83.46

Medium Transmission line model (150 km)



$\eta_{\text{no-loss}}$	98.85
$\eta_{\text{after-loss}}$	91.64

It can be noticed that as distances increase, the additional losses affect the efficiency significantly. As this basic principle is evident and confirmed via the results obtained, this GUI section works effectively too.

C. SEVERITY INDEX

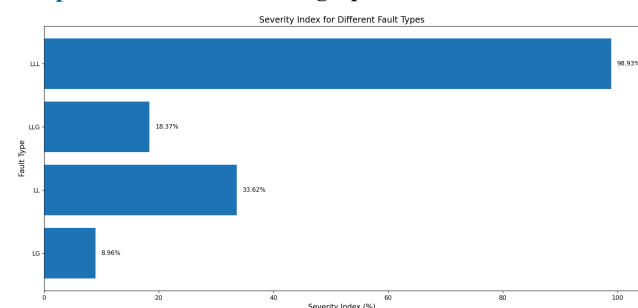
Table 6: Simulink Parameters used while calculating the severity index for a short transmission line model

R (Resistance Value)	7 Ω		
L (Inductance Value)	0.1945 H		
Efficiency (normal condition)	93.53%		
Fault Time	20 s		
Stop Time	2-5 s		
Fault Block Parameters	$R_{\text{FAULT}} = 0.001 \Omega$	$R_{\text{GROUND}} = 0.01 \Omega$	$C_{\text{SNUB}} = \text{inf}$

Table 7: Severity Index of a short transmission line for various faults

S No	Fault Type	Fault-efficiency (%)	Severity Index (%)
1	LG	85.15	8.96
2	LL	60.28	33.62
3	LLG	76.37	18.37
4	LLL	1.0032	98.93

Graph: Table 7 inferred as a graph



V. CONCLUSION

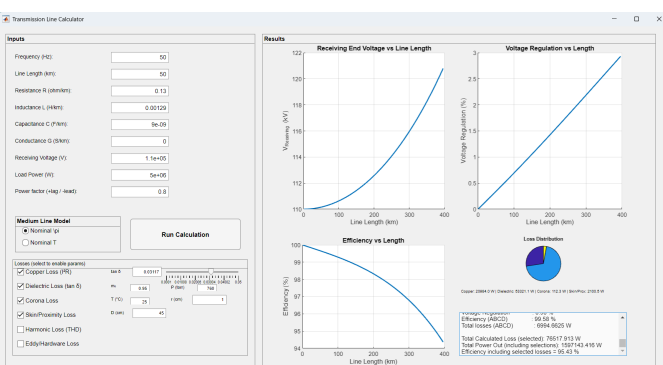


FIGURE 3: Overall working GUI

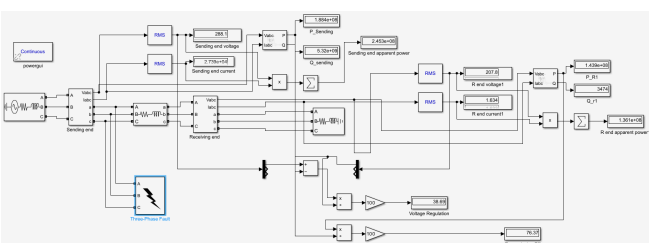


FIGURE 4: Simulink Model of the short transmission line

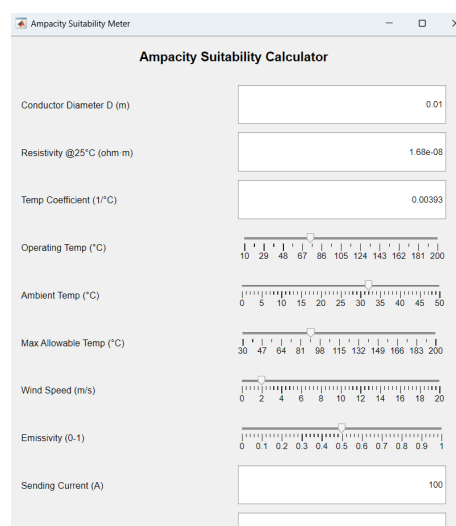


FIGURE 5: Ampacity Score Calculation

MATLAB is an excellent software when it comes to numerical calculations, control optimization, signal processing, but when it is used as a full-scale app development platform, it is not as efficient as it would usually be. It is a single-threaded UI, when the code runs, the GUI cannot update, plots lag and computations take a lot of time. That's why a full-scale app integration is not done but every individual aspect's results have been validated. The same idea can be extended and integrated with the help of modern day platforms like JavaScript.

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